SIVA PRASANNA

Software Engineer | Java Full Stack Development | Oracle PL/SQL

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SUMMARY

Results-driven Software Engineer with 6+ years of IT experience at Tata Consultancy Services (TCS), combining Java Full Stack Development with enterprise Oracle PL/SQL database engineering. Proficient in Java 21, Spring Boot, Spring Security, RESTful APIs, MongoDB Atlas, and responsive frontend development using HTML5, CSS3, and JavaScript. Skilled in high-volume batch processing, API security (JWT with token rotation), database query tuning, Docker containerization, and automated testing with JUnit 5 and Mockito. Experienced in Agile delivery, legacy modernization, and end-to-end enterprise application support.

SKILLS

Languages: Java 21, PL/SQL, SQL, JavaScript, HTML5, CSS3

Frameworks & Core Technologies: Spring Boot, Spring Data JPA, Spring Security 6, Hibernate ORM, RESTful Web Services, MVC Architecture, Clean Architecture

Databases & Cloud: Oracle Database 19c, MongoDB Atlas, Snowflake

Database Engineering: Stored Procedures, Packages, Functions, Triggers, Views, Cursors, Query Optimization, Performance Tuning, Bulk Collect & FORALL

DevOps, Web Servers & Tools: Docker, Docker Compose, Nginx, Git, GitHub, Jenkins (CI/CD), Maven, Postman, Swagger / OpenAPI 3, Control-M, PeopleTools

Security & Testing: JWT Authentication & Token Rotation, Role-Based Access Control (RBAC), JUnit 5, Mockito, MockMvc, Automated Unit & Integration Testing, Agile/Scrum, SDLC

PROJECTS

CareerNest - Enterprise Full-Stack Job Portal Platform

Tech Stack: *Java 21, Spring Boot, MongoDB Atlas, Spring Security 6, JWT (Token Rotation), Twilio Cloud SMS, HTML5, CSS3, JavaScript, Docker, Docker Compose, Nginx, Render*

- Architected and developed a full-stack job portal connecting employers and job seekers, integrating a decoupled Spring Boot REST API with an interactive, responsive frontend built using **HTML5, CSS3, and modular Vanilla JavaScript**.
- Built dynamic frontend user interfaces including real-time job discovery with instant keyword/location filtering, a 1-click modal application workflow, a candidate application tracker with live status badges (`APPLIED`, `UNDER_REVIEW`, `ACCEPTED`, `REJECTED`), and an employer KPI metrics review board.
- Engineered a lightweight, zero-build client architecture with custom CSS design tokens (Flexbox/Grid), modular JavaScript Fetch API client (`api.js`) with automatic Bearer token injection, and production static hosting on Nginx Alpine.
- Implemented an enterprise security architecture with Spring Security and stateless JJWT, featuring 15-minute access tokens and single-use 7-day refresh tokens with automated token rotation and revocation in MongoDB Atlas to eliminate replay vulnerabilities.
- Integrated a dual-mode Twilio Cloud SMS notification engine to asynchronously broadcast real-time status updates directly to candidate mobile devices upon employer review actions.
- Engineered high-performance MongoDB Atlas schemas with compound unique indexing (`jobId + seekerId`) to guarantee application idempotency, preventing duplicate candidate submissions and optimizing filtering queries.
- Containerized the full-stack system using multi-stage Docker builds (Temurin JDK 21 backend + Nginx Alpine frontend) and Docker Compose, achieving live production deployment on Render Cloud with OpenAPI 3 (Swagger UI) documentation.

E-Commerce Backend System API

Tech Stack: *Java, Spring Boot, MongoDB, Spring Security, JWT, AWS EC2, Render, Docker*

- Architected a production-ready E-Commerce REST API using Java (Spring Boot) and MongoDB, featuring stateless JWT authentication (JJWT) and Role-Based Access Control (RBAC) across USER, ADMIN, and MANAGER roles with BCrypt password hashing.
- Engineered zero-lock concurrency control using MongoDB conditional atomic updates (findAndModify) to prevent race conditions, stock overselling, and inventory inconsistencies during simultaneous checkouts.
- Designed a 4-step transactional order workflow (stock validation -> atomic inventory deduction -> order creation -> cart clearing) with historical price locking (priceAtPurchase) and automated stock replenishment upon order cancellation.
- Implemented an idempotent payment simulation service with safe retry policies, ensuring network retries or duplicate client requests never double-charge or corrupt order states.
- Optimized query performance and schema design by balancing document referencing (User-Order) with sub-document embedding (Cart-CartItems), supporting dynamic pagination and category filtering with Spring Data Pageable.
- Deployed live environments on AWS EC2 and Render, integrating SpringDoc OpenAPI 3 / Swagger UI for interactive documentation and @RestControllerAdvice for RFC-compliant global exception handling.

EXPERIENCE

TATA CONSULTANCY SERVICES (TCS) | Enterprise Solutions - Colruyt HR & Payroll

Senior Systems Engineer | Oct 2019 - Present

PRR Time and Attendance | PL/SQL Developer | Mar 2026 - Present

- Lead the migration of employee time-and-attendance processes from a mainframe environment to Oracle PL/SQL programs.
- Support the generation of PeopleSoft queries used by third-party agencies to validate employee absence and working hours.
- Manage the end-to-end process, including post-fix support, performance improvements, and program-efficiency enhancements.

Tax Reduction for Logistics Workers | Project Lead & PL/SQL Developer | Nov 2024 - Feb 2026

- Worked with clients to gather requirements and define business rules for identifying logistics workers eligible for tax reductions.
- Analyzed large datasets and developed PL/SQL procedures to extract, filter, and process employee data against government eligibility criteria.
- Created PeopleSoft components in Oracle HCM to display and manage eligible workers' tax details and supported unit and system testing.

Decommissioning of Legacy Payroll | PL/SQL Developer | Mar 2024 - Oct 2024

- Identified and mapped obsolete PeopleSoft pages, components, and backend logic for decommissioning using PL/SQL and PeopleTools.
- Decommissioned legacy cycles involving PL/SQL programs, SQRs, CTLs, and SQL scripts.
- Supported transition to PeopleSoft 9.2 Core HCM by safely retiring outdated payroll processes.

C-People Linked to Payroll | PL/SQL & PeopleSoft Developer | Oct 2022 - Mar 2024

- Migrated payroll and provisions components from PeopleSoft 7.5 to PeopleSoft 9.2 Core HCM.
- Converted SQR programs to PL/SQL for interim/job-student payroll, including automated verification emails.
- Resolved payroll discrepancies by developing front-end pages and backend PeopleCode logic to meet client requirements.
- Developed PL/SQL programs and Control-M schedules for employee payroll files supporting DIMONA, DMFA, and Social Passive processes.

Social Passief to Finance | Project Lead & Developer | Mar 2022 - Oct 2022

- Designed and developed PL/SQL programs and Control-M cycles to calculate refund amounts returned to company profits when employees left service during a given year, including end-to-end logic and file generation.

Payslips of Payroll | PeopleTools & PL/SQL Developer | May 2021 - Mar 2022

- Developed payroll payslip components in PeopleTools Application Designer and implemented backend business logic using PeopleCode.

Light Projects | PeopleSoft Developer | Oct 2020 - May 2021

- Delivered maintenance, enhancement, and legal-compliance changes to PeopleSoft pages and components.
- Optimized SQL queries in PL/SQL programs to improve data retrieval speed and overall program efficiency.

KEY ACHIEVEMENTS

- Recognized as Best Performer of the Year for converting complex SQR programs to PL/SQL for payroll calculations and improving performance in C-People projects.
- Received client appreciation for excellence, dedication, loyalty, and contributions across onsite and offshore assignments.
- Contributed across the delivery lifecycle, including AS-IS analysis, solution design, development, system testing, integration testing coordination, regression testing, defect fixes, and hypercare support.
- Successfully delivered two major projects and multiple medium, small, maintenance, enhancement, and production-support initiatives.
- Best Performance Award for the JOFO project for exceptional performance within short span of time.

CERTIFICATIONS

- Claude Certified Developer - Foundation (2026)
- Executive Java Full Stack Developer Certification (2026) - HCL/GUVI Zen Class (IIT Madras Research Park)
- SnowPro Associate Certification (2025)
- OCI 2025 AI Foundation Certification (2025)

EDUCATION

- Bachelor of Engineering (Mechanical Engineering) | PSNA College of Engineering and Technology, Dindigul | 2019
- Higher Secondary (12th Standard) | Shri Renuga Vidhyalaya Matriculation Hr. Sec. School, Theni, Tamil Nadu | 2015
- SSLC (10th Standard) | Shri Renuga Vidhyalaya Matriculation Hr. Sec. School, Theni, Tamil Nadu | 2013